

HOANG VIET DO

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Research Interests

Applied harmonic analysis; wavelets and multiscale methods; geometry of data; graph theory and combinatorics.

Education

Yale University Ph.D. Student in Mathematics	2025–present
Hanoi Institute of Mathematics, Vietnam Master's Student in Mathematics	2022–2024
VNUHCM–University of Science, Vietnam B.S. in Mathematics	2022

Research Experience

Applied Harmonic Analysis and Multiscale Methods Working with Prof. Ronald Coifman. On wavelets, multiresolution methods, data geometry, and the Questionnaire framework.	2026–present
Littlewood–Offord Problems and Combinatorial Statistics Research with Tuan Tran, Hoi H. Nguyen, and Van H. Vu.	2024–2025
Combinatorial Commutative Algebra Research on ordinary and symbolic powers of fiber products of ideals in polynomial rings. Supported by the Vietnam National Foundation for Science and Technology Development (NAFOSTED).	2023–2024
Spectral Graph Theory, Fractal and Hyperbolic Geometry Pre-Graduate Fellowship in Mathematics; supervised by Prof. Thomas Koberda.	2022–2024

Publications and Preprints

Littlewood–Offord bounds on the symmetric groups and applications. With Hoi H. Nguyen, Kiet H. Phan, Tuan Tran, and Van H. Vu. 2025, <i>preprint</i> .
On the ordinary and symbolic powers of fiber products. With Hop D. Nguyen and Seyed Amin Seyed Fakhari. <i>Journal of Algebra</i> 678 (2025), 570–600.
Intersection graphs of general linear groups. With M. H. Bien. <i>Journal of Algebra and its Applications</i> 20 (2021), 2150039.
Group identities on units of division rings. With M. Ramezan-Nassab and M. H. Bien. <i>Communications in Algebra</i> 49 (2021), 3010–3019.

Selected Honors

- **Bronze Medal, International Mathematical Olympiad**, 2018.
- **Vingroup Master's Scholarship**, 2022 and 2023.
- **VNUHCM Prize for Highly Qualified Papers**, 2022.
- **VIASM Scholarship for Prominent Undergraduates in Mathematics**, 2019 and 2021.

Selected Academic Experience

Mathematics Graduate Student Seminar, Yale University Organizer and Speaker	Fall 2026
Projects in Mathematics and Applications, Vietnam Mentor; prepared tests and lectures and interviewed candidates.	Summer 2024
VNUHCM–University of Science, Vietnam Instructor for the university mathematics olympiad team; lectures on algebra and combinatorics problem solving.	Spring 2023